Qiskit v2.x Practice Quiz

Questions

1. What does layout refer to in transpilation?

    A. Mapping virtual qubits to physical qubits
    B. Choosing the number of shots
    C. Drawing the circuit
    D. Choosing classical register names

2. Why may SWAP gates be inserted during transpilation?

    A. To reset qubits
    B. To satisfy hardware connectivity constraints
    C. To increase shot count
    D. To remove all noise

3. OpenQASM is best described as:

    A. A language for representing quantum programs
    B. A physical quantum processor
    C. A Python plotting library
    D. A classical database system

4. What does the Sampler primitive primarily return?

    A. Expectation values
    B. Measurement samples or bitstring results
    C. Only statevectors
    D. Only transpiled circuits

5. Which state lies on the positive X axis of the Bloch sphere?

    A. $|0\rangle$
    B. $|1\rangle$
    C. $|+\rangle$
    D. $|-\rangle$

6. Which of these is not a valid Estimator PUB entry?

    A. a single parametarized quantum circuit

B. $asinglequantumcircuitthatcontainsnoparameters$
B. target precision
C. number of shots

7. What does the Bloch sphere represent?

A. Only classical bits
B. Single-qubit pure states
C. Only two-qubit entangled states
D. Classical probability distributions only

8. What does increasing the number of shots usually improve?

A. Sampling precision
B. Circuit depth
C. Qubit connectivity
D. Gate fidelity automatically

9. What is the result of applying Z to $|1\rangle$?

A. $|0\rangle$
B. $|1\rangle$
C. $-|1\rangle$
D. $|+\rangle$

10. What is the purpose of transpilation?

A. To convert a circuit into backend-compatible instructions
B. To measure all qubits automatically
C. To create a statevector
D. To define an observable

11. When is Sampler more appropriate than Estimator?

A. When you need bitstring measurement outcomes
B. When you need expectation values of observables
C. When you need to draw a circuit
D. When you need to create a backend

12. Which of the following passes are used in heuristic stage of transpilation layout stage? *(Select all that apply)*

A. TrivialLayout
B. DenseLayout
C. VF2Layout
D. SabreLayout

13. Which gate maps $|0\rangle$ to $|+\rangle$?

A. X

B. Z

C. H

D. CX

14. Which OpenQASM 3 variable declaration corresponds to Qiskit's ClassicalRegister declaration?

    A. bit

    B. clreq[ ]

    C. bit[ ]

    D. clbit[ ]

15. A Qiskit Runtime Estimator job is executed with precision = 0.02. The precision value is then changed to 0.01, while all other settings remain unchanged. Which of the following best describes the expected effect?

    A. Fewer shots are used

    B. Precision only affects Sampler, not Estimator

    C. target precision

    D. Backend changes automatically to the Sampler

16. Given the code below, how would you retrieve probability distribution from sampler results? Assume all components have been imported correctly.

```
qbits = QuantumRegister(2,"qbits")
clbits = ClassicalRegister(2,"clbits")
qc = QuantumCircuit(qbits, clbits)
(q0, q1) = qbits
(c0, c1) = clbits
qc.h(0)
qc.cx(0,1)
qc.measure_all()
service = QiskitRuntimeService()
backend = service.least_busy()
pm = generate_preset_pass_manager(backend=backend, optimization_level=2)
qc_isa = pm.run(qc)
sampler = SamplerV2(mode=backend)
job =sampler.run([qc_isa])
result = job.result()
```

    A. counts = result[0].data.meas.get_counts()

    B. counts = result[0].data.clbits.get_counts()

    C. counts = result[0].data.evs.get_counts()

    D. counts = result[0].data.stds.get_counts()

17. Which of the following is used for optimization considering code below? Assume all prerequisties were imported.

```
qc = QuantumCircuit(10)
qc.h(0)
for i in range(9):
    qc.cx(i, i+1)
service = QiskitRuntimeService()
backend = service.least_busy()
pm = generate_preset_pass_manager(2, backend=backend)
qc_isa = pm.run(qc)
```

A. TrivialLayout

B. CXCancellation

C. CommutativeCancellation

D. KAK Decomposition

18. What does an X gate do to $|0\rangle$?

A. Creates $|+\rangle$

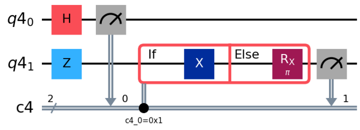
B. Maps it to $|1\rangle$

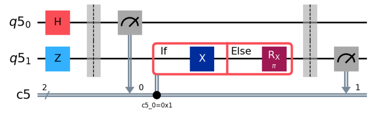
C. Measures the qubit

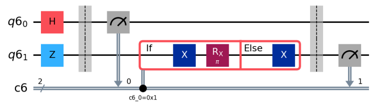
D. Adds a global phase only

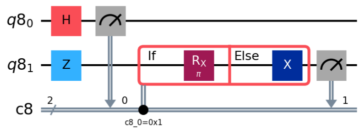
19. Which one of the following images is the output from the code below:

```
from qiskit import *
from math import pi

qubits = QuantumRegister(2)
clbits = ClassicalRegister(2)
qc = QuantumCircuit(qubits, clbits)
(q0, q1) = qubits
(c1, c2) = clbits

qc.h(0)
qc.z(1)
qc.barrier()
qc.measure(q0, c1)
with qc.if_test((c1,1)) as else_:
    qc.x(q1)
with else_:
    qc.rx(pi, q1)
qc.barrier()
qc.measure(q1,c2)
qc.draw(output="mpl", plot_barriers=False)
```

A. 

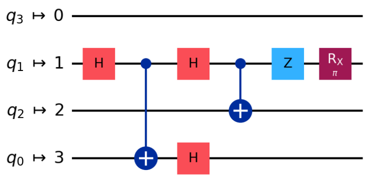
B. 

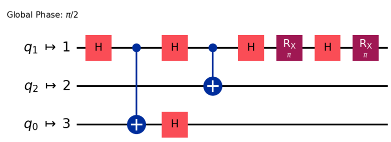
C. 

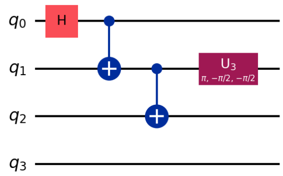
D. 

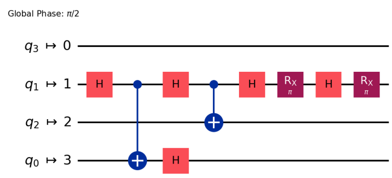
20. Which of the circuits below is possible output for the code below?

```
from qiskit import *
from qiskit_ibm_runtime import QiskitRuntimeService
from qiskit.transpiler.preset_passmanagers import generate_preset_pass_manag
from math import pi

qc = QuantumCircuit(4)
qc.h(0)
for i in range(2):
    qc.cx(i,i+1)
qc.z(1)
qc.rx(pi,1)
service = QiskitRuntimeService()
backend = service.least_busy()
pm = generate_preset_pass_manager(
    optimization_level=1,
    basis_gates=["rx","cx","h"],
    coupling_map=[[0,2],[1,3],[1,2]]
)
qc_isa = pm.run(qc)
qc_isa.draw("mpl",idle_wires=True)
```

A. 

B. 

C. 

D. 

Answer Key

1. S4-003: A
2. S4-002: B
3. S8-001: A
4. S5-001: B
5. S2-002: C
6. S6-005: D
7. S2-001: B
8. S7-001: A
9. S1-003: C
10. S4-001: A
11. S5-002: A
12. S3-004: B, D
13. S1-002: C
14. S8-003: C
15. S6-008: B
16. S7-005: A
17. S3-005: C
18. S1-001: B
19. S3-003: A
20. S3-006: D